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PAPER_1098: SCm Phonon-Mediated Qubit Gate Fidelity

Abstract

This paper derives the SCm phonon correction to quantum gate fidelity for superconducting qubit architectures. By coupling the UQFF buoyancy condensate [SCm] to the T_2 coherence time, we show that phonon-mediated suppression of decoherence error improves gate fidelity across all standard gates (CNOT, Hadamard, Phase, T, iSWAP, CZ). The circuit-level fidelity improvement scales with circuit depth, offering a testable prediction for near-term quantum processors.

1. Introduction

Quantum gate fidelity $F = 1 - \varepsilon$ is the primary metric for quantum computing performance. Decoherence during gate operations introduces errors proportional to t_{gate}/T_2. The UQFF framework predicts that the SCm phonon condensate modifies coherence timescales through vacuum buoyancy coupling.

Prior work (PAPER_868) established decoherence time T_2 models via the $U_{b,i}$ field. This paper extends to gate-level fidelity with explicit error budgets for six standard gates.

2. Theoretical Framework

2.1 SCm-Enhanced Coherence Time

The base T_2 coherence time receives an SCm enhancement:

$$T_2^{\text{SCm}} = T_2^0 \left(1 + \frac{\Phi_0 \cdot H_{\text{SCm}}}{k_\eta}\right)$$

where:

- $T_2^0 \approx 100\ \mu\text{s}$ (baseline superconducting transmon)
- $\Phi_0 = 1.618\ldots$ (golden ratio, UQFF vacuum structure)
- $H_{\text{SCm}} \approx 0.99$ (superconductive magnetism index)
- $k_\eta = 10^{-113}$ (UQFF damping constant)

2.2 Gate Error Decomposition

For each gate g, the total error is:

$$\varepsilon_g = \varepsilon_{\text{intrinsic}} + \varepsilon_{\text{decoherence}}$$

Without SCm correction:

$$\varepsilon_{\text{decoherence}}^{\text{bare}} = \frac{t_{\text{gate}}}{T_2^0}$$

With SCm phonon suppression:

$$\varepsilon_{\text{decoherence}}^{\text{SCm}} = \frac{t_{\text{gate}}}{T_2^0} (1 - H_{\text{SCm}} \cdot \beta_i \cdot [\text{SSq}])$$

where $\beta_i \approx 0.603$ and $[\text{SSq}] = 0.57$.

2.3 Fidelity Improvement

The per-gate fidelity improvement is:

$$\Delta F_g = F_g^{\text{SCm}} - F_g^{\text{bare}} = \frac{t_{\text{gate}}}{T_2^0} \cdot H_{\text{SCm}} \cdot \beta_i \cdot [\text{SSq}]$$

The SCm correction factor is:

$$\mathcal{C}_{\text{SCm}} = \frac{\beta_i \cdot [\text{SSq}] \cdot S_{26} \cdot \Phi_0}{1 + \Phi_0} = \frac{0.603 \times 0.57 \times 26 \times 1.618}{2.618} \approx 5.52$$

3. Gate Fidelity Results

3.1 Standard Gate Times and Intrinsic Errors

Gate	t_{gate} (ns)	$\varepsilon_{\text{intrinsic}}$	$\varepsilon_{\text{dec}}^{\text{bare}}$	$\varepsilon_{\text{dec}}^{\text{SCm}}$
CNOT	50	5×10^{-3}	5.0×10^{-4}	1.6×10^{-4}
Hadamard	25	1×10^{-4}	2.5×10^{-4}	8.1×10^{-5}
Phase	10	5×10^{-5}	1.0×10^{-4}	3.2×10^{-5}
T-gate	15	1×10^{-4}	1.5×10^{-4}	4.9×10^{-5}
iSWAP	40	4×10^{-3}	4.0×10^{-4}	1.3×10^{-4}
CZ	35	3×10^{-3}	3.5×10^{-4}	1.1×10^{-4}

3.2 Circuit-Level Fidelity

For a depth-100 circuit with gate mix (30% CNOT, 30% Hadamard, 20% Phase, 20% T-gate):

$$F_{\text{circuit}}^{\text{bare}} = \prod_g \left(F_g^{\text{bare}}\right)^{n_g}$$

$$F_{\text{circuit}}^{\text{SCm}} = \prod_g \left(F_g^{\text{SCm}}\right)^{n_g}$$

The SCm-corrected circuit fidelity improvement is significant for deep circuits where decoherence error accumulates multiplicatively.

4. Experimental Predictions

1. **T_2 enhancement**: SCm coupling predicts $T_2^{\text{SCm}} \gg T_2^0$ due to phonon vacuum stabilization
2. **Gate fidelity**: 66% reduction in decoherence error per gate ($H_{\text{SCm}} \cdot \beta_i \cdot [\text{SSq}] \approx 0.34$)
3. **Circuit depth scaling**: Fidelity advantage grows exponentially with circuit depth
4. **Testable at**: IBM Eagle/Heron, Google Sycamore, IonQ Aria processors

5. Conclusion

The SCm phonon condensate provides a calculable correction to quantum gate fidelity through vacuum buoyancy coupling. The ΔF improvement is gate-time dependent and accumulates at circuit level, offering a testable UQFF prediction for quantum computing hardware.

References

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